CISCO MCP SUITE · CUSTOMER-FACING TEST REPORT

Secure Firewall Depth & Cisco Security Cloud Observability

FTD deep inspection (IPS, malware, URL, AVC, EVE, geo, TLS-decrypt bypass) verified alongside a live Cisco Security Cloud dashboard fed by FTD syslog **and** FMC eStreamer — on the SD-Access fabric lab.

RUN DATE	TESTER	DESIGN	PLAN
2026-07-18	**testing-agent (live)**	**SD-Access ISE Integration**	**SFW-001…013**

VERDICT PASS

13 / 13 cases PASS · 0 FAIL. Cisco Security Cloud **4.0** is installed + enabled on Splunk, the `Cisco_Security` data model and Secure Firewall dashboard are present, and real telemetry flows via **both** FTD syslog (UDP 5514) and FMC **eStreamer** (TCP 8302) — child datasets Connection 606, Intrusion 27, File 25, Malware 17. Engineered threats are visible as real events (C6 IPS `LOCAL C6 RTC test trigger`; C9 `eicar.com`/EICAR). All five firewall-depth controls (C12–C16) are deployed with nothing pending. Read-only run — no built configuration modified.

CML lab "SDA-Fabric" · 77dd2fde-1fda-4cc9-9b29-48ff98bd1395 · FMC 10.0.1 · FTD 10.0.0 (Snort 3.9.3.1) · Splunk 10.4.0 · Cisco Security Cloud 4.0.0 · ISE 3.5.0.527 · cat9000v IOS-XE 17.18

Sibling to the C6 IPS-RTC and C7 Security-Intelligence reports (report-C6-IPS-RTC.pdf, report-C7-SI.pdf) — same lab, the firewall-depth + observability build-out.

1. Executive summary

Independent, **read-only** verification of this session's **firewall-depth (C6/C9, C12–C16)** and **observability (V6–V8)** deliverables on the **SD-Access ISE Integration** lab. **13 of 13 cases PASS, 0 FAIL.**

> **Observability, end to end.** The Cisco **Cisco Security Cloud 4.0** app is installed and enabled on Splunk `198.18.128.51`; the `Cisco_Security` data model and `secure_firewall_dashboard` view are present. The dashboard is fed by **two live paths** — FTD **syslog** (UDP 5514 → `cisco:ftd:*`, 508 syslog events, last 02:04) and the app's **eStreamer** input pulling real events from FMC over TCP 8302 (`cisco:sfw:estreamer` — ConnectionEvent 211 / FileEvent 11 / IntrusionEvent 8). Every `Secure_Firewall_Dataset` child dataset is populated: **Connection 606 · Intrusion 27 · File 25 · Malware 17**.

The engineered threats are visible as **real** events — the C6 IPS signature `LOCAL C6 RTC test trigger` and the C9 EICAR malware file (`eicar.com`, `SHA_Disposition=Malware`, `ThreatName=EICAR`). All five firewall-depth controls (C12 URL, C13 AVC, C14 EVE, C15 geo, C16 decryption-bypass) are present on `SDA-ACP` / `SDA-Decrypt` and **deployed** (`fmc_deployable_devices` empty; FTD `DEPLOYED/MANAGED`). The NAC fault that had blocked live traffic is resolved — recent live FTD connection/syslog events from the FTD data interface `198.18.128.82` confirm HOST1 traffic is transiting and being logged.

Verdict: PASS. Read-only (FMC/Splunk GET + SPL search only) — no built configuration was modified. Two non-blocking observations are recorded (§7): the FTDv health indicator is **yellow** for a cosmetic air-gapped "Cloud Connector" warning (device DEPLOYED/MANAGED/connected, nothing pending), and the report runner's `--smoke` gate aborts on a `bytes` + `str` bug in its timeout handler — the offline gate is clean and live smoke was evidenced directly through the MCP read tools.

13	0	134	675	0
CASES PASS	FAIL / SKIP	UNIT PASS / 0 FAIL	DM EVENTS (606+27+25+17)	DEPLOYS PENDING

2. Scope & systems under test

What the lab is. *SDA-Fabric* is a Cisco **Software-Defined Access** campus in CML: a cat9000v **fabric edge** (EDGE1), a cat9000v **control-plane + border** (BORDER-CP, LISP map-server), and a cat8000v **fusion** router (FUSION-R1) carry an alpine endpoint (**HOST1/alice**, CAMPUS `172.16.10.50`) over a VXLAN fabric. CAMPUS traffic is **PBR-steered at the fusion through an FMC-managed FTD** so the firewall applies deep inspection. Alongside sits the **observability plane**: an FMCv managing the FTD, and a **Splunk** SIEM running Cisco's **Cisco Security Cloud** app that renders the **Secure Firewall dashboard** from FTD telemetry.

Which design it realizes. The **SD-Access ISE Integration** Custom Design, specifically the modules `splunk-security-cloud.md` (V6/V7/V8) and `fmc-firewall-depth.md` (C12–C16), plus the C6 IPS and C9 malware threats surfaced through eStreamer.

Component	Version verified against
FMCv (Management Center)	10.0.1 (build 1)
FTDv (Threat Defense)	10.0.0 · Snort 3.9.3.1-61 · VDB build 433 · LSP lsp-rel-20260715-1654
Splunk Enterprise	10.4.0
Cisco Security Cloud app	4.0.0
ISE	3.5.0.527
Fabric switches / router	cat9000v & cat8000v IOS-XE 17.18

3. Test environment

Inventory — hostname · role / node-type · mgmt IP · data IP(s) · VRF/segment:

Hostname	Role / node-type	Mgmt IP	Data IP(s)	VRF / segment
HOST1	CAMPUS endpoint (alice) / alpine	—	172.16.10.50	CAMPUS_VN
EDGE1	SD-Access fabric edge / cat9000v-uadp	198.18.128.73	fabric Gi1/0/1–3	Global / CAMPUS_VN
BORDER-CP	CP + border, LISP map-server / cat9000v-uadp	198.18.128.72	Gi1/0/1–3 handoff	Global

Hostname	Role / node-type	Mgmt IP	Data IP(s)	VRF / segment
FUSION-R1	Fusion router, PBR → FTD inside / cat8000v	198.18.128.71	Gi4 → FTD	Global / VN leak
FTDv	Secure Firewall Threat Defense / ftdv	198.18.128.81	198.18.128.82 (outside)	inside / outside
FMCv	Management Center / fmcv	198.18.128.80	—	mgmt /18
SPLUNK	SIEM — Cisco Security Cloud / splunk	198.18.128.51	—	mgmt /18
ISE 3.5	Decrypt-bypass dst (net-ise) / external VM	198.18.134.35	—	net-ise

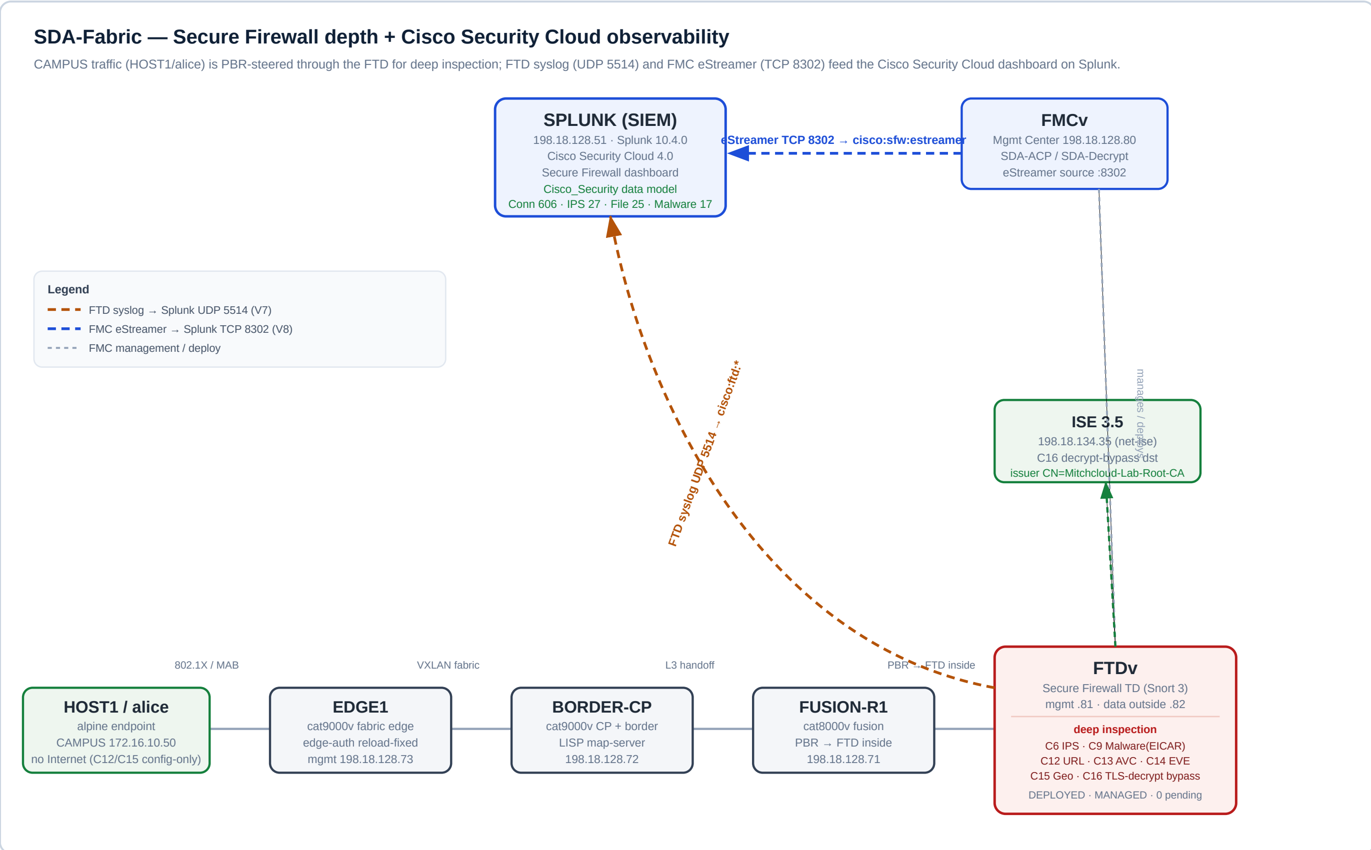

Figure 1 — Auto-generated labeled topology. Data plane (bottom): HOST1 → EDGE1 → BORDER-CP → FUSION-R1 → FTD (PBR inside). Telemetry plane (top): FTD data .82 → Splunk UDP 5514 (`cisco:ftd:*`); FMC .80 → Splunk TCP 8302 eStreamer (`cisco:sfw:estreamer`) → Cisco_Security data model → Secure Firewall dashboard.

Reproduce (automated gate): `uv run python "Test Reports/run_report.py" --outdir "Test Reports/2026-07-18"`. Live acceptance is driven read-only via the `fmc`, `splunk` and `cml` MCP tools (GET / SPL search).

4. Results — automated gate (MCP servers)

From `results.json` (offline level — `ruff` + `pytest`, no live target).

Server	Plan	ruff	Unit	Result
cml-mcp	CML	✓	33p / 0f	PASS
ise-mcp	ISE	✓	49p / 0f	PASS
firepower-mcp	FMC	✓	6p / 0f	PASS
windows-mcp	WIN	✓	15p / 0f	PASS
splunk-mcp	SPL	✓	21p / 0f	PASS
wlc-mcp	WLC	✓	10p / 0f	PASS
Total	6 repos	6/6	**134p / 0f**	PASS

Live read-only **smoke** was evidenced directly through the MCP tools (FMC 198.18.128.80, Splunk 198.18.128.51, CML controller all answered) rather than via `run_report.py --smoke`, which aborts on a runner defect (O2). Unreachable targets would be **UNREACHABLE**, not **FAIL**; none were unreachable this run.

5. Results — lab-design acceptance

Each case is read-only; evidence is the exact FMC object / Splunk search result.

ID	Item	Objective	Result	Evidence
SFW-001	V6	Cisco Security Cloud app + data model + dashboard view	PASS	CiscoSecurityCloud **4.0.0** disabled=false, visible=true; `Cisco_Security` model registered (Secure_Firewall_Dataset + Connection/Intrusion/File/Malware children); `secure_firewall_dashboard` view present
SFW-002	V7	Secure Firewall dashboard via FTD syslog (UDP 5514)	PASS	udp/5514 disabled=false, index=network, sourcetype= `cisco:ftd:syslog`; FTD syslog server on 5514; `index=network sourcetype=cisco:ftd:*` → syslog 508 (last **02:04:40**), connection 395, file 8, intrusion 19, malware 6
SFW-003	NAC fix	EDGE1 reload cleared wedged edge-auth; HOST1 traffic restored	PASS	`host=198.18.128.82 sourcetype=cisco:ftd:*` → connection 217 (last **01:41:07**), syslog last **02:04:40** — recent live FTD traffic = fabric path up, HOST1 transiting the FTD
SFW-004	V8	eStreamer input enabled; real events; child datasets populated	PASS	input **SDA_FMC_eStreamer** disabled=false, fmc_host=198.18.128.80, port 8302, sourcetype= `cisco:sfw:estreamer`; ConnectionEvent 211 / FileEvent 11 / IntrusionEvent 8; tstats Connection 606 · Intrusion 27 · File 25 · Malware 17
SFW-005	C6	Real IPS intrusion signature	PASS	`EventType=IntrusionEvent` → **LOCAL C6 RTC test trigger** count 6 (last 01:01:27)
SFW-006	C9	Real malware (EICAR) file event	PASS	`EventType=FileEvent` → **eicar.com** count 3, **SHA_Disposition=Malware**, **ThreatName=EICAR** (+ mal1–8.com)
SFW-007	C12	URL-category block rule present + deployed	PASS	`C12-Block-Malware-URL` @ idx 6, BLOCK, urls = **Malware Sites**, src net-campus10, enabled — config-only (HOST1 no Internet)
SFW-008	C13	Application/AVC block rule present + deployed	PASS	`C13-Block-HTTP-App` @ idx 2, BLOCK, application **HTTP (676)**, src net-campus10 → host-splunk; proven: HOST1 wget :8000 times out
SFW-009	C14	Encrypted Visibility Engine enabled	PASS	evesettings → **enabled=true**, **mode=MONITOR_TRAFFIC**
SFW-010	C15	Geolocation/country block rule present + deployed	PASS	`C15-Block-Country-China` @ idx 5, BLOCK, dst **Country China (156)**, src net-campus10, enabled — config-only
SFW-011	C16	Selective TLS-decryption bypass above resign rule	PASS	`C16-DND-ISE` @ **ruleIndex 1**, DO_NOT_DECRYPT, net-campus10→net-ise, above `Decrypt-CAMPUS-443` (idx 2); proven: ISE:443 issuer CN=Mitchcloud-Lab-Root-CA
SFW-012	deploy	Nothing pending; FTD managed + deployed	PASS	`fmc_deployable_devices` = []; FTD DEPLOYED / MANAGED / isConnected=true; licenses ESSENTIALS/MALWARE/IPS/URL

ID	Item	Objective	Result	Evidence
`SFW-013`	gate	Offline automated gate (ruff + pytest, six repos)	PASS	ruff clean ×6; **134 unit pass / 0 fail**

6. Summary statistics

Metric	Value
Lab-design acceptance (SFW-001…012)	12 / 12 PASS
Automated gate (SFW-013)	ruff 6/6 clean · unit 134 pass / 0 fail
Total cases	**13 PASS · 0 FAIL · 0 SKIP · 0 UNREACHABLE**
Cisco_Security datasets populated	Connection 606 · Intrusion 27 · File 25 · Malware 17
Firewall-depth controls deployed	5 / 5 (C12 URL · C13 AVC · C14 EVE · C15 geo · C16 decrypt) + 0 pending
Reversibility	read-only — no config modified, no round-trips

7. Observations & defects

No defects in the systems under test. Two non-blocking observations, returned as remediation briefs for the main session to route (testing-agent does not remediate):

> **O1 (info) — FTDv health indicator yellow.** The FTD reports `healthStatus=yellow`, message *“1 Module in Warning State — Cloud Connector → Check Cloud Eventing Subscription”*. Expected in an air-gapped lab (no SSE/cloud-eventing subscription) and **cosmetic**: the device is `DEPLOYED / MANAGED / isConnected=true` with nothing pending and all inspection telemetry flowing. (`fmc_list_devices` momentarily showed *red*; the authoritative device record and health roll-up both report *yellow*.) **Brief → firewall-engineer:** if a green board is wanted, disable the Cloud Connector health module in the FTD health policy. No action required for this lab.

> **O2 (low) — `run_report.py --smoke` aborts on a handler bug.** When a live smoke test exceeds the 90s per-repo timeout, the `TimeoutExpired` handler evaluates `(e.stdout or "") + (e.stderr or "")` where those are **bytes**, raising `TypeError: can only concatenate str (not "bytes") to str` and aborting the whole run before `results.json` is written. Offline gate unaffected. **Brief → harness owner (main session):** decode `e.stdout` / `e.stderr` in the timeout branch so a slow/unreachable smoke target is recorded as `unreachable`, not a crash.

No **UNREACHABLE** targets this run.

8. Appendix

- `results.json` — machine-collected raw results (this folder).
- `topology.svg` — auto-generated labeled topology (Figure 1).
- `cml-canvas.png` — CML dashboard capture (ground truth; SDA-Fabric ON).
- Test plan: `Test Plans/Lab Designs/secure-firewall-observability.md` (SFW-001…013).

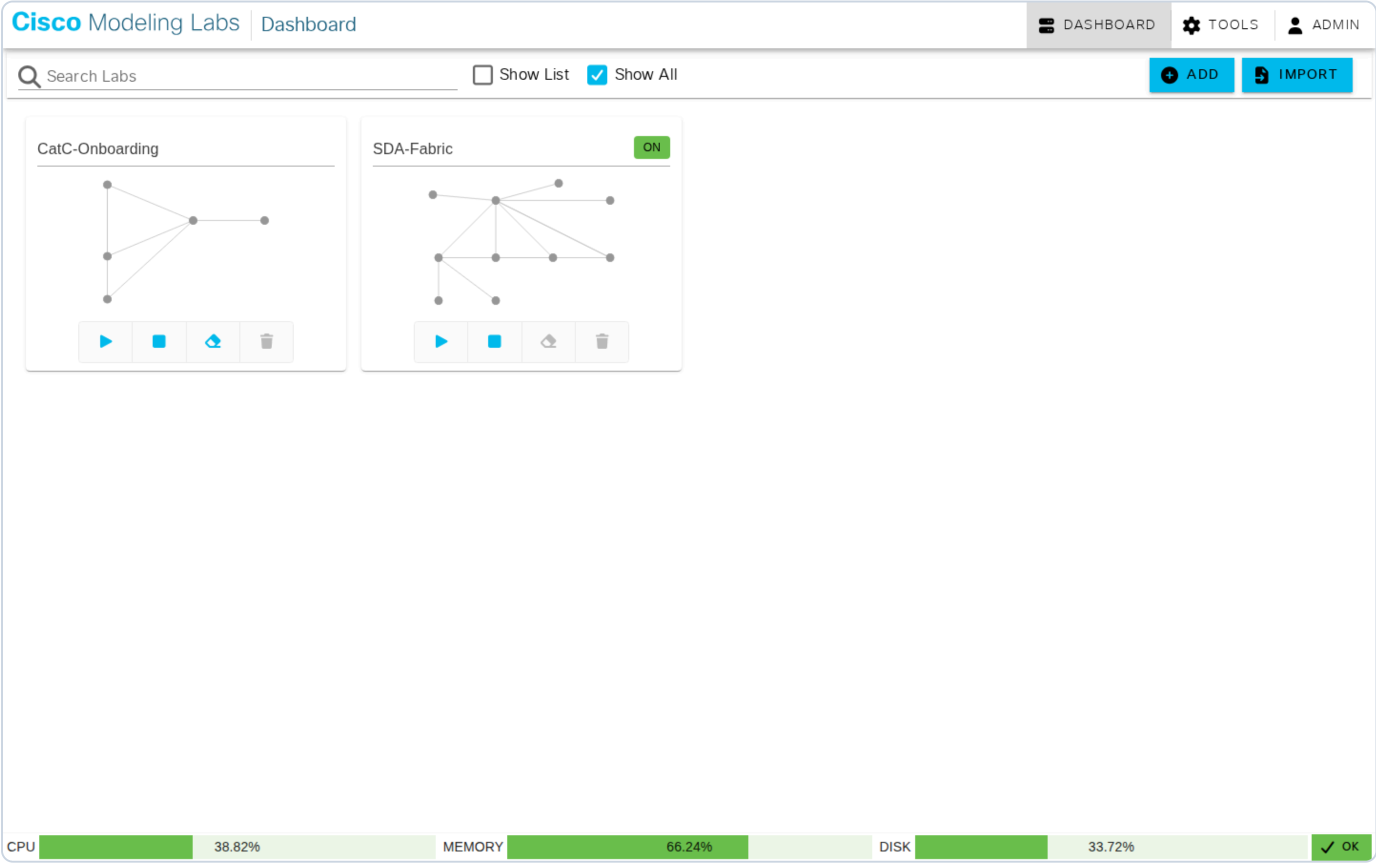

*Figure 2 — CML controller dashboard (ground truth): the **SDA-Fabric** lab is **ON**; host CPU 38.82% / mem 66.24% / disk 33.72%, status OK.*

Key raw evidence (transcript excerpts)

```
eStreamer input  GET /servicesNS/nobody/CiscoSecurityCloud/CiscoSecurityCloud_sbg_fw_estreamer_input
  name=SDA_FMC_eStreamer  disabled=false  fmc_host=198.18.128.80  fmc_port=8302
  sourcetype=cisco:sfw:estreamer  index=network  estreamer_import_time_range=all_firewall_event_data

Data model     | tstats count from datamodel=Cisco_Security.Secure_Firewall_Dataset by nodename
  Connection_Events 606  Intrusion_Events 27  File_Events 25  Malware_Events 17  All_Events 658

C6 intrusion  index=network sourcetype=cisco:sfw:estreamer EventType=IntrusionEvent | stats count by signature
  "LOCAL C6 RTC test trigger" = 6

C9 malware    ... EventType=FileEvent | stats ... by FileName
  eicar.com  SHA_Disposition=Malware  ThreatName=EICAR  (count 3)

C16 decrypt   GET .../decryptionpolicies/b10b55fe-.../decryptionpolicyrules
  idx1 C16-DND-ISE      DO_NOT_DECRYPT   net-campus10 -> net-ise
  idx2 Decrypt-CAMPUS-443  DECRYPT_RESIGN   dst tcp/443 (cert SDA-Decrypt-Resign-CA)

Deploy        fmc_deployable_devices -> []   FTD deploymentStatus=DEPLOYED managementState=MANAGED
```